Md Miadul Islam Nizzan

Computer Science Undergraduate | AI/ML, NLP & AI-Driven Software

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PROFILE

Computer Science and Engineering undergraduate with project experience spanning artificial intelligence, machine learning, natural language processing, computer vision, multi-agent systems, software engineering, networking, cybersecurity, and embedded systems. Combines research-oriented development with student mentoring, IEEE leadership, technical documentation, and collaborative program delivery.

EDUCATION

Green University of Bangladesh — Expected January 2027
B.Sc. in Computer Science and Engineering · Rupganj, Narayanganj, Bangladesh

- Current CGPA: 3.70 / 4.00
- Dean's Award for outstanding academic performance
- Vice Chancellor's Award for academic excellence

Royal Media College — 2021
Higher Secondary Certificate (HSC) · Mymensingh, Bangladesh

Shaheen Cadet School — 2019
Secondary School Certificate (SSC) · Bhaluka, Mymensingh, Bangladesh

RESEARCH

Robustness, Explainability, and Attribution Stability of Lightweight Vision Transformers for Federated Chest X-ray Classification under Non-IID Client Distributions
Undergraduate Thesis · Ongoing, 2026 – present
A federated binary-classification study for pleural effusion detection using NIH ChestX-ray14 across 10 simulated non-IID hospital clients.

- Comparing centralized, local-only, FedAvg, and FedProx training with DeiT-Tiny and ResNet-18
- Noisy-label robustness and attribution-stability analysis
- Grad-CAM and Vision Transformer attribution methods to evaluate explanation consistency across client distributions

Robustness and Tokenization Stability of Transformer Models for 12-Class Bangla Dialect Classification Under Character-Level Perturbations
Manuscript · Submitted — decision pending
Evaluated three Transformer models on 59,453 BanglaDial records under progressive character-level perturbations.

- Multilingual DistilBERT was most robust at the 20% perturbation condition
- Typo-aware augmentation improved its accuracy from 51.005% to 62.913%

Multilevel Detection of Social Media Harassment Targeting Women
Research in Progress · Dataset development and text-classification model fine-tuning
Curating a multilevel dataset of social-media harassment targeting women and developing a structured labeling scheme for model training and evaluation.

- Multilevel dataset curation and a structured labeling scheme
- Fine-tuning a text-classification model
- Preparing the study for a future research manuscript, subject to experimental results

TECHNICAL EXPERTISE

AI, Machine Learning & NLP	TensorFlow, Keras, PyTorch, Hugging Face Transformers, scikit-learn, LangChain, RAG pipelines, Multi-agent workflows, CNNs, MobileNetV2, BERT-based NLP, Bangla NLP
Agentic AI	LangChain, Groq API, Tavily search, Tool calling, Role-specific prompting, Source attribution, Streamlit deployment
Federated & Explainable AI	FedAvg, FedProx, Non-IID learning, DeiT-Tiny, ResNet-18, Grad-CAM, Vision Transformer attribution
Programming	Python, JavaScript, TypeScript, Dart, PHP, SQL, C (working knowledge), C++ (working knowledge), Java (working knowledge), x86 Assembly (working knowledge)
Web & Mobile	React, Vite, Node.js, Django REST Framework, Flask, Flutter, REST APIs, HTML/CSS
Data & Databases	Pandas, NumPy, OpenCV, MySQL, PostgreSQL, SQLite, Supabase

Security & Networking	TCP/IP, Packet analysis, AES, RSA, SHA-256, Network fundamentals, Cryptographic verification
Tools	Git, GitHub, Jupyter, Google Colab, VS Code, NetBeans, XAMPP, LaTeX, Microsoft Office

SELECTED PROJECTS

AI Research Assistant Agent GitHub

Python, LangChain, Groq, Tavily, Streamlit

- Web-search tool calling with role-specific prompts for each agent
- Explicit failure handling and saved reports

Smart Farming System: Bengali-First Agricultural Platform GitHub

React, TypeScript, Vite, Node.js, PostgreSQL

- Plant-disease detection, automated irrigation control, and expert consultation workflows
- Market-price verification and multi-provider AI chat

CogniSearch: Research-Oriented AI Chatbot

Python, Streamlit, Groq API, Llama 3.3

- Temperature, Top-P, and token controls with persistent conversation state
- Markdown export and secure credential configuration

Fosholer Bondhu: AI Agricultural Assistant

Python, TensorFlow, Keras, MobileNetV2, TensorFlow Lite

- MobileNetV2 transfer-learning workflow for disease classification
- Structured for evaluation and TensorFlow Lite deployment on-device

Farm Operation & Transaction Management GitHub

Flutter, Django REST Framework, JWT, SQLite, PostgreSQL

- Offline-first data model for use in low-connectivity settings
- End-to-end coverage from animal health records to financial reporting

Secure File Sharing & Cryptographic Verification GitHub

Python, Flask, AES-128-CBC, RSA, SHA-256

- Cryptographic primitives implemented from first principles, not library calls
- Role-based access control with tamper simulation and audit logging

EXPERIENCE & LEADERSHIP

Vice Chair · IEEE GUB Student Branch, Executive Committee 2026 – Present

Coordinate technical events, workshops, student initiatives, communications, and cross-team collaboration.

Student Mentor · Green University of Bangladesh 2025 – Present

Guide junior students in academic planning, technical coursework, projects, GitHub, presentations, and documentation.

Assistant Graphic Designer · IEEE GUB Student Branch, Executive Committee 2025 – 2026

Supported branding, event promotion, technical program publicity, and digital content development.

Assistant Teacher · Kolatoli Darus Sunnah Al Arabiah Madrashah 2023 – 2024

Conducted structured lessons, mentored students, and developed instructional communication and classroom leadership skills.

AWARDS, MEMBERSHIP & SERVICE

- **Vice Chancellor's Award** · Green University of Bangladesh
- **Dean's Award** · Green University of Bangladesh
- **Debate Champion** · Royal Media College (2020)
- **IEEE Member** · IEEE (2025)
- **IEEE Day Ambassador** · IEEE GUB Student Branch
- **STI 2025 Conference Volunteer** · IEEE (2025)